Chujun Zong

Dr.-Ing. · Life Cycle Assessment · Sustainable Building Design

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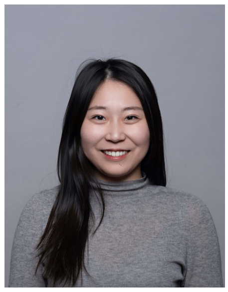

RESEARCH PROFILE

Dr.-Ing. (magna cum laude, TUM 2026) specializing in dynamic life cycle assessment of the built environment. Developer of two original computational frameworks, GraphDLCA (graph-database-driven dynamic LCA) and MOSO (multi-objective stochastic optimization under uncertainty), with 20+ peer-reviewed publications including four first-author articles in JCR Q1 journals (Automation in Construction, Building and Environment, Scientific Reports). Co-managed two federally funded research projects (BMWK, DFG), supervised 18 Master's theses, and received the Doce et Delecta Teaching Award (First Place). Trilingual in English, German, and Mandarin.

RESEARCH INTERESTS

Dynamic life cycle assessment | Multi-objective stochastic optimization under uncertainty | Graph-database modeling for building inventories | Early-stage design decision support for sustainable building | EU sustainability regulation.

EDUCATION

Dr.-Ing. Technical University of Munich (TUM) **2022 – 2026**

Chair of Energy Efficient and Sustainable Design and Building. Supervisor: Prof. Werner Lang. magna cum laude.

- Dissertation: Addressing Future Changes in Building Life Cycle Assessment: Uncertainty Modeling and Dynamic Life Cycle Assessment (DLCA) for Decision-Making.

M.Sc. Resource Efficient and Sustainable Building, Technical University of Munich (TUM) **2019 – 2021**

Passed with Merit.

- Thesis: Implementation of a Stochastic Methodology to Predict Window Opening Behavior in a Dynamic Building Simulation Program (published and presented at BauSim 2022).

B.A. Architecture, Technical University of Munich (TUM) **2014 – 2019**

Exchange B.A. Architecture, École Polytechnique Fédérale de Lausanne (EPFL) **2016 – 2017**

PROFESSIONAL EXPERIENCE

Research Fellow & Doctoral Candidate — Technical University of Munich **2022 – 2026**

Chair of Energy Efficient and Sustainable Design and Building (Prof. Werner Lang).

- Co-managed two federally funded research projects (BMWK CircularGreenSimCity, DFG EarlyBIM2), responsible for LCA methodology, web tool development, workshop organization, deliverables and reporting within interdisciplinary consortia.
- Developed and published two computational frameworks: GraphDLCA (Neo4j graph database for automated dynamic LCA, Automation in Construction 2026) and MOSO (multi-objective stochastic optimization, Building and Environment 2022/2024).
- Supervised 18 Master's theses (16 completed), four of which led to peer-reviewed conference publications with students as authors.
- Taught four courses across English and German curricula (ca. 333 contact hours), receiving the department teaching award (First Place, 2023).
- Maintained the chair's web presence.

- Coordinated communications and institutional liaison activities with partner universities, including the South China University of Technology and TUM Asia.
- Contributed to the chair's recruitment of research associates.
- Mentored over five student research assistants (HiWi) across teaching, research projects and administrative work.

Working Student — Siemens Energy AG — 2020 – 2021

Section of Construction, Design & Sustainability in Real Estate.

- Developed internal CO_2 pricing methodology and drafted the first corporate sustainability and workplace guidelines since the corporate restructuring.
- Provided guidance on sustainable factory renovation, in collaboration with factory site managers and project managers on international projects.
- Developed sustainable and agile working space renovation plans for offices in Germany, Singapore, Malaysia and Slovenia, including CAD drawing, material and furniture selection, and solution presentation.
- Presentations to the managing board and input sessions within Real Estate department.

Independent consulting engagement on F-gas regulation and CBAM — 2026 – now

- Advisory work on EU F-gas Regulation and CBAM compliance for industrial clients.

FUNDED RESEARCH PROJECTS

CircularGreenSimCity — 2022 – 2025

Holistic life cycle assessment of urban districts. Funded by the German Federal Ministry for Economic Affairs and Climate Action (BMWK).

- Responsible for the LCA work package of archetype-based district stock modeling, scenario development, assessment methodology, and the project's web tool programming.
- Embedded and supervised five Master's theses within the project, with results feeding deliverables and three conference publications.
- Responsible for milestone and final reporting, budget management, consortium workshop organization across partners of university, municipality and consulting company.
- Mentored four student research assistants.

EarlyBIM2 — 2022 – 2024

Knowledge database and machine learning assistance for performance-oriented building. Funded by the German Research Foundation (DFG).

- Developed early-design assessment methods under incomplete information, including the benchmarking of machine-learning surrogates against physics-based simulation.
- Responsible for milestone and final reporting, external communication with participating universities, regular onsite presentations, and collaborative publications (four joint publications in total, including the consortium's final joint publication on adaptive detailing strategies).
- Initiated the open-source publication of knowledge base EarlyData for wider public use.
- Mentored one student research assistant.

TEACHING EXPERIENCE

Student Supervision

18 Master's theses supervised (2023 – 2026, 16 completed, 2 ongoing), spanning dynamic LCA, building energy modeling, machine learning, and multi-objective optimization. Four theses led directly to peer-reviewed international conference publications with students as authors, with two further journal

manuscripts in preparation. Topics anchored in funded projects, doctoral research, and industry cooperation (e.g., Siemens Energy). One thesis grew into a student start-up.

System Effects and Interdependencies of Sustainable Planning in Civil Engineering (Master, interdisciplinary seminar, English) **2025 – 2026**

Co-lecturer, Technical University of Munich

Full responsibility for own seminar group. Ca. 17 contact hours, ca. 50 students per semester.

Introduction to Scientific Writing (Bachelor and Master, seminar, English) **2023 – 2026**

Sole lecturer, Technical University of Munich

Full responsibility. Ca. 97 contact hours, ca. 40 students per semester.

Sustainable Building & Ecological Building Basic Modules (Bachelor, lecture and exercise, German) **2022 – 2026**

Lead lecturer (exercises), teaching assistant (lectures), Technical University of Munich

Main responsibility. Ca. 82 contact hours, ca. 200 students per semester. Mentored 1-2 student research assistants per semester.

Sustainable Lighting Technology (Master, lecture, English) **2022 – 2025**

Sole lecturer, Technical University of Munich

Full responsibility. Ca. 137 contact hours, ca. 100 students per semester. Doce et Delecta Teaching Award (First Place), 2023. Mentored 1 student research assistant per semester.

HONORS & AWARDS

- Doce et Delecta Teaching Award, First Place — Department of Civil and Environmental Engineering, TUM (Winter Semester 2022/2023, Sustainable Lighting Technology)
- National Award for Outstanding Self-funded Students Abroad (Category A), 2024 – the national distinction for Chinese self-funded doctoral researchers abroad.

• ACADEMIC INVOLVEMENT & SERVICES

- Peer reviewer for international journals including Building and Environment, Building Simulation, Buildings, Sustainability, Journal of Architectural Engineering and Design Management, Scientific Reports, Discover Artificial Intelligence, Cell Reports Sustainability, etc.
- Presentations at international conferences including BauSIM (2022), Building Simulation (2023, 2025), and WSBE (2024).
- Peer reviewer for international conferences including Building Simulation (2025) and WSBE (2024).
- Member of IBPSA(-DACH)
- DGNB Registered Professional

RESEARCH SOFTWARE & TECHNICAL COMPETENCES

- GraphDLCA: graph-database structure and calculation framework for automated dynamic LCA of buildings (Neo4j/Cypher, Python, demonstration video on my personal homepage).
- MOSO / MOSO-II: multi-objective stochastic optimization framework coupling building energy simulation with embodied carbon objectives (NSGA-II, Monte Carlo, Python).
- Contributions to the CircularGreenSimCity district assessment web tool (Python).
- Open-source publication of EarlyData knowledge base from EarlyBIM2 (Zenodo: https://doi.org/10.5281/zenodo.10962087).
- Other competences: MATLAB, building simulation (e.g., IDA ICE, EnergyPlus, TRNSYS), architectural design (e.g., AutoCAD, Adobe Creative Suite, ArchiCAD, Revit)
- Languages: Chinese (native), German (professional level), English (professional level, IELTS 8.0), French (conversational level).

SELECTED PUBLICATIONS

Peer-Reviewed Journal Articles

- **Zong, C.**, Banihashemi, F., Petzold, F. & Lang, W. (2026). Graph-based Database Structure and Calculation Framework for Automated Dynamic Life Cycle Assessment of Buildings. Accepted for publication in Automation in Construction. (JCR Q1)
- **Zong, C.**, Banihashemi, F., & Lang, W. (2025). Dynamic life cycle impact assessment (DLCIA) in a sustainable building planning process. Scientific Reports, 15(1), 32680. (JCR Q1)
- **Zong, C.**, Chen, X., Deghim, F., Staudt, J., Geyer, P., & Lang, W. (2024). A holistic two-stage decision-making methodology for passive and active building design strategies under uncertainty. Building and Environment, 251. (JCR Q1)
- **Zong, C.**, Margesin, M., Staudt, J., Deghim, F., & Lang, W. (2022). Decision-making under uncertainty in the early phase of building façade design based on multi-objective stochastic optimization. Building and Environment, 226, 109729. (JCR Q1)
- Napps, D., Staudt, J., Saluz, U., Chen, X., Steiner, D., **Zong, C.**, Deghim, F., Lang, W., Geyer, P., Schnellenbach-Held, M., Petzold, F., Borrmann, A., König, M. (2026). Evaluation of Building Design Variants in Early Phases on the Basis of Adaptive Detailing Strategies. Buildings, 16(4), 685. (JCR Q2)
- Banihashemi, F., Weber, M., Deghim, F., **Zong, C.**, & Lang, W. (2024). Occupancy modeling on non-intrusive indoor environmental data through machine learning. Building and Environment, 254. (JCR Q1)

Selected Conference Papers

- **Zong, C.**, & Lang, W. (2025). Dynamic factors in life cycle assessment: systematic review and categorization. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Lang, W., Pancheri, F. & Sun, Y. (2025). Assessing pedestrian level barriers using mobile robot to validate the 15-minute city concept. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Sun, Y., & Lang, W. (2024). System Dynamics Modeling of Life Cycle Carbon Footprints for Building Wall Insulation Materials. IOP Conference Series: Earth and Environmental Science, 1363.
- **Zong, C.**, Reitberger, R., Deghim, F., Staudt, J., Larkin, J. A., & Lang, W. (2023). A comparative study of dynamic building simulation and machine-learning within a two-stage multi-objective stochastic optimization framework. Building Simulation Conference Proceedings, 18, 1540–1547.
- **Zong, C.**, Banihashemi, F., Vollmer, M., & Lang, W. (2022). Implementation of occupant behaviour models for window control using co-simulation approach. BauSIM 2022.